

Education

- Class 2025 **B.Sc. Computer Science & Statistics**, *University of Toronto*, Toronto, ON, Honours.
- Teaching Assistant: **CSC369 Operating Systems** | Graduate Coursework: **CSC413 Neural Networks & Deep Learning**
- Audit **Deep Learning Systems**, *Carnegie Mellon University*.

Technical Skills

- Programming Python, C, C++, CUDA, Rust, Java, TypeScript, SQL
- Frameworks PyTorch, TensorRT-LLM, vLLM, Triton, DDP, FSDP, DeepSpeed ZeRO, RLHF Linux, Docker

Professional Experience

- 2026 **Software Engineer, Pre-AI Training (Contractor)**, *LinkedIn*, Remote.
- Designed and executed **LLM evaluation frameworks** for code generation (Python, TypeScript, SQL); engineered adversarial test cases for correctness, edge-case robustness, and efficiency—outputs fed directly into LinkedIn's **LLM fine-tuning** and **RLHF** pipeline, improving model alignment on software engineering tasks.
 - Curated **instruction-following datasets** via structured **prompt engineering**: designed paired prompt-completion examples, applied systematic code-quality rubrics, and produced high-signal **supervised fine-tuning (SFT)** data across reasoning, debugging, and code synthesis domains.
- 2025 **Software Engineer**, *Ontario Teachers' Association*, Toronto, ON.
- Architected end-to-end **ML pipeline observability** using Prometheus and Grafana for a platform serving 18k+ concurrent users; established latency SLOs, automated regression detection, and built **real-time data processing** dashboards reducing incident MTTR by **35%**.
 - Engineered **Docker**-based CI/CD provisioning and health-check automation, cutting deployment cycle time **40%**; applied reproducible environment configs mirroring **ML experiment reproducibility** practices across distributed infrastructure.
- 2023 **Systems Engineer**, *RBC Capital Markets*, New York City.
- Profiled and optimized latency-sensitive **data pipelines** in trading risk/pricing infrastructure; reworked **caching layers** and fetch paths on P&L-sensitive flows, achieving **70% end-to-end latency reduction**—directly applicable to low-latency **LLM inference serving**.
- 2019–2020 **Software Engineer**, *IBM Watson AI*, Markham, CA.
- Contributed to **Watson Commerce Insights**, an enterprise **AI/ML analytics** platform; built CI/CD test infrastructure and integration suites that reduced production regressions across **ML-powered** release cycles.

Selected Projects

- 2025 **LLM Inference Optimization: Serving Infrastructure & Kernel Optimization**, *TensorRT-LLM · Triton · CUDA · PyTorch · vLLM*.
- Benchmarked **TensorRT-LLM** with **INT8** weight-only quantization and in-flight batching against vLLM across batch sizes 1–64; applied **FlashAttention-2** and **FP8** to attention layers on A100; authored fused **softmax + dropout** kernel in **Triton**, improving memory bandwidth from **61% to 89%** of peak via Nsight Compute.
- 2024 **Deep Learning Engine & GPU Kernel Optimization**, *C++ · CUDA · Triton · Nsight · Python*.
- CUDA Kernel**: Tiled shared-memory matmul reaching **14.2 TFLOP/s** (87% of cuBLAS peak), **823 GB/s** bandwidth—**3.1×** over naive; roofline analysis guided register tiling optimizations.
 - CNN + Transformer**: Conv (forward & backward), ResNet9 on CIFAR-10; multi-head self- and cross-attention (pre-norm) encoder/decoder stack from scratch, matching reference perplexity within **0.3%**.
- 2025 **Foundation Model Pretraining: LLM Training, Kernel Optimization & RLHF**, *PyTorch · Triton · FSDP · CUDA · Python*.
- Built end-to-end pretraining pipeline from raw **Common Crawl** (BPE tokenizer, quality filtering, deduplication) into a **124M-parameter** pre-norm Transformer (RoPE, SwiGLU, GQA) trained with AdamW + cosine LR; matched **GPT-2 small perplexity within 1.8%** at **60% of published compute budget** via Chinchilla-optimal scaling laws; implemented **FlashAttention-2** from scratch in **Triton** (**2.1×** over PyTorch SDPA, 84% A100 bandwidth) and scaled to 4×A100s with **FSDP** ZeRO-3, **DDP**, gradient checkpointing, and **BF16**—**3.2×** throughput, **41%** peak memory reduction vs. single-GPU baseline.